```
> cat incident-report-lab12.md
```

# Incident Playbook: Reclaiming the Fleet

Lab 12: Fleet Patching with AWS Systems Manager

**Environment:** NorthBank Prod | **Module:** AWS Day 2 Operations & Troubleshooting | **Status:** RESOLVED

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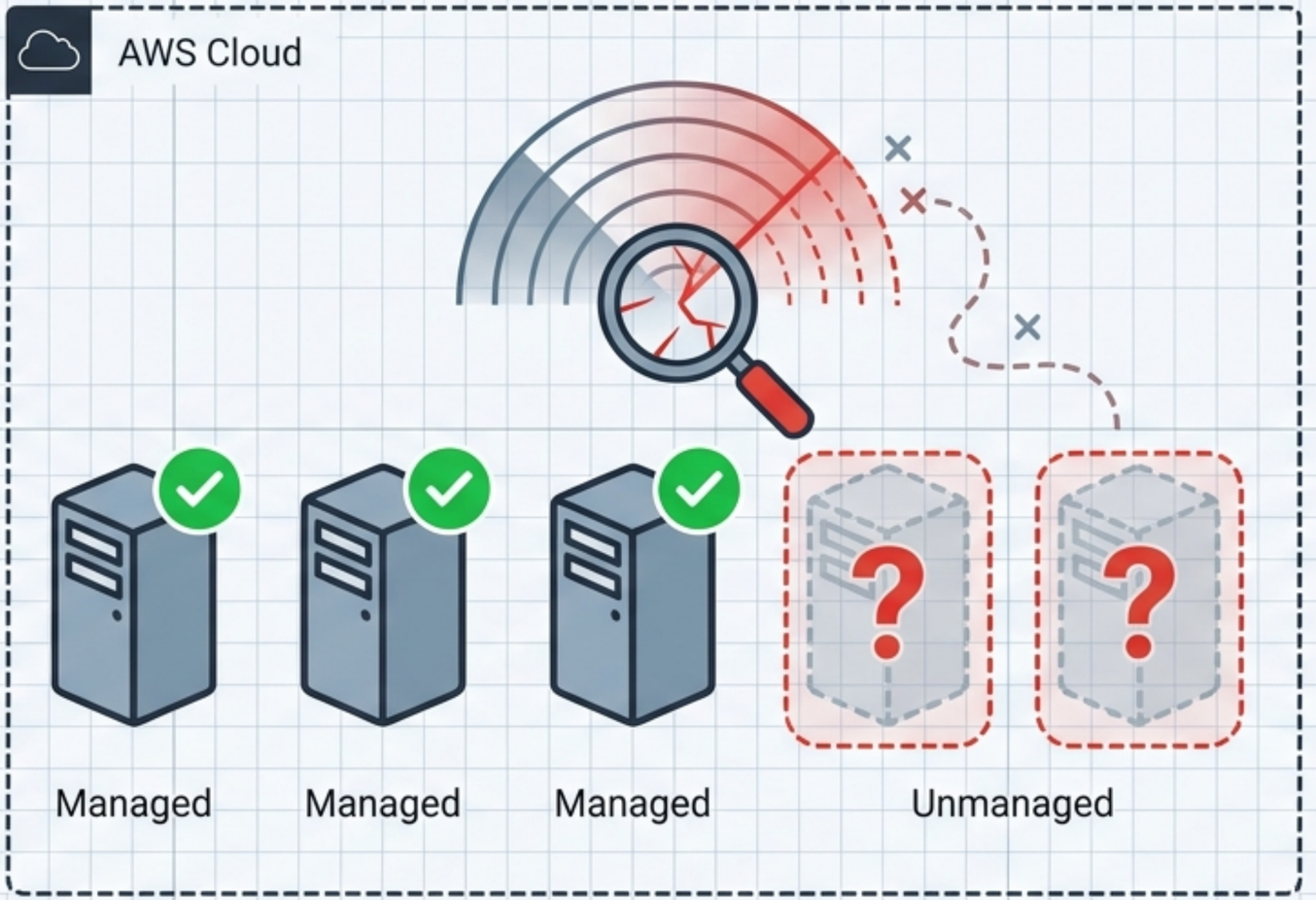
# Ticketing System

**Priority: SEV-3**

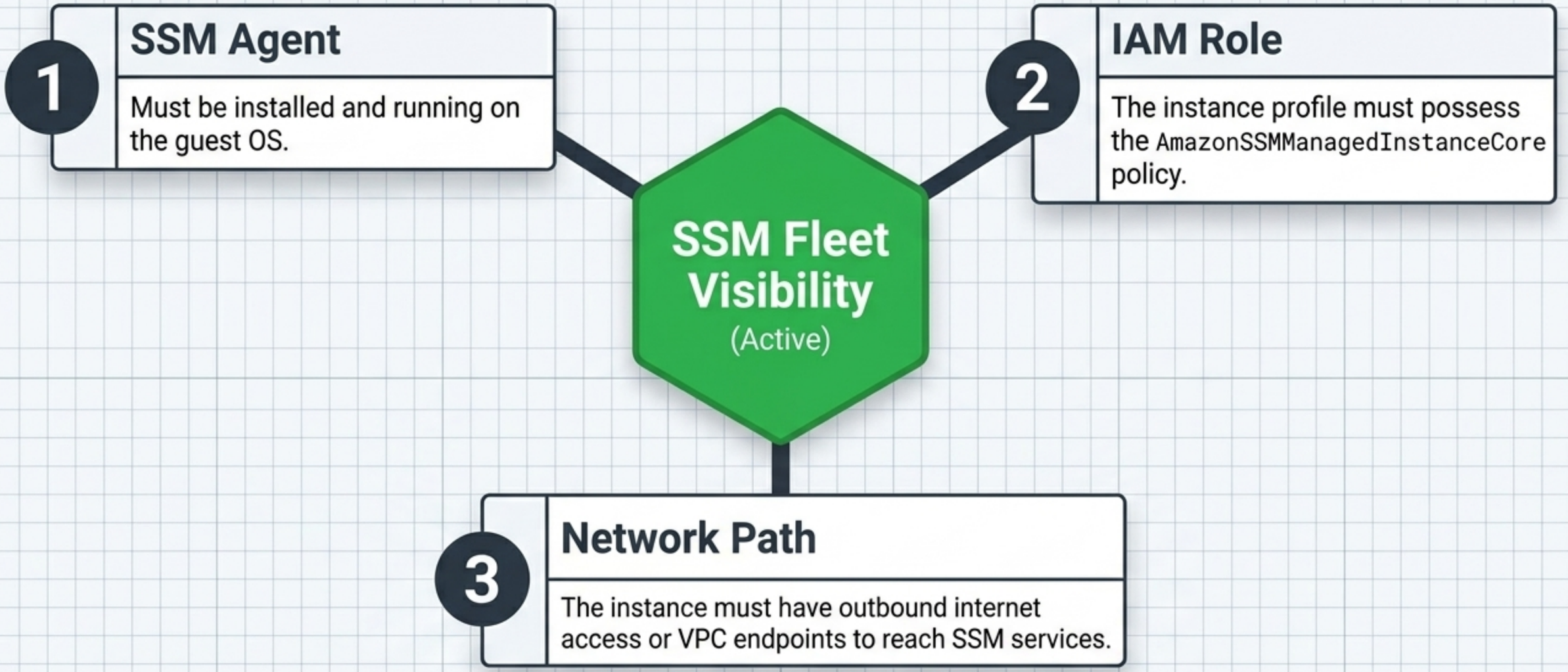
| | |
|---|---|
| **Reporter:** | **Aisha (Security/GRC)** |

**Issue Description:** Patch report shows missing patches. Some instances vanished from SSM entirely.

# Visibility Gap




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**Diagnostic Rule:** An instance will vanish from `describe-instance-information` if any of these three prerequisites fail.

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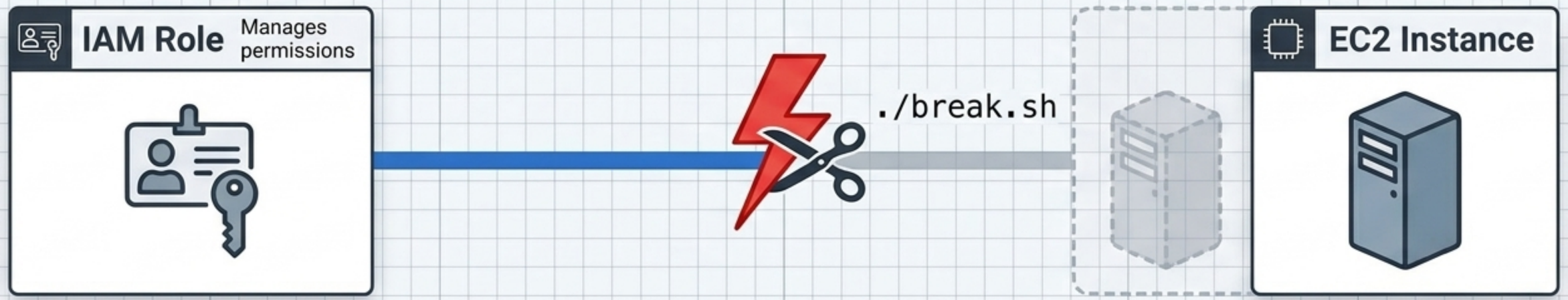
# CLI Execution Timeline

**Action:** Severing the connection

```
$ export ROLE_NAME, ASG_NAME from terraform output
$ ./break.sh
$ aws iam list-attached-role-policies --role-name $ROLE_NAME
[ ]
```

The SSM policy is gone.

# Architectural State Change




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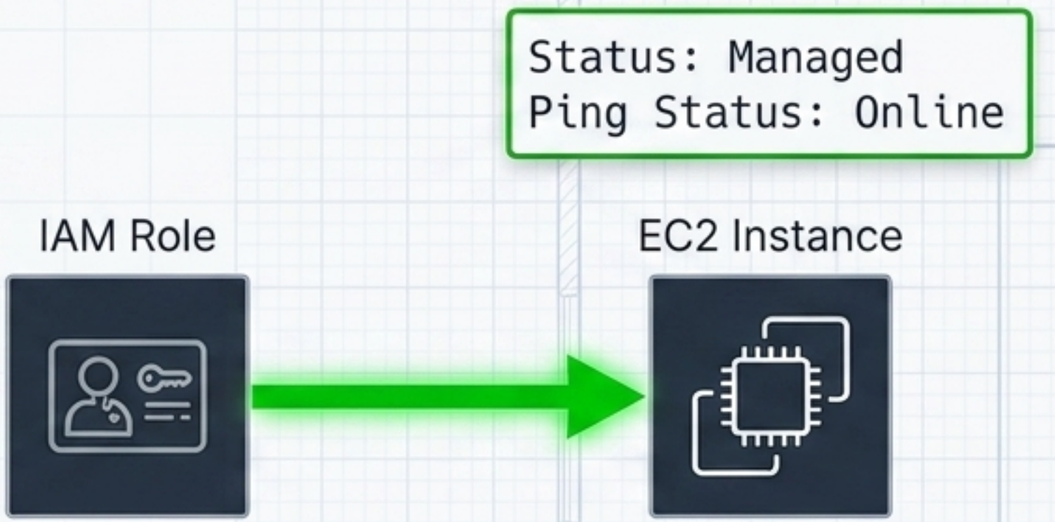
Action: Restoring the policy

Verification

```
$ aws iam attach-role-policy --
role-name $ROLE_NAME --policy-arn
arn:aws:iam::aws:policy/AmazonSSMM
anagedInstanceCore

$ aws ssm describe-instance-
information

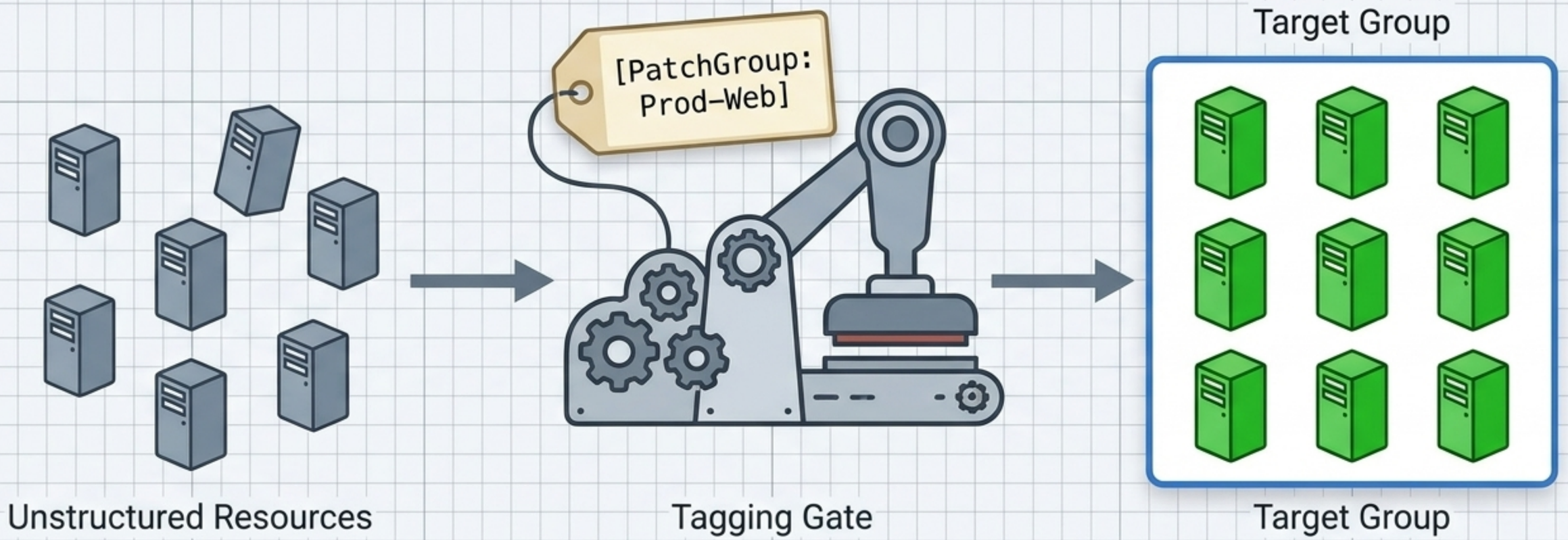
$
```




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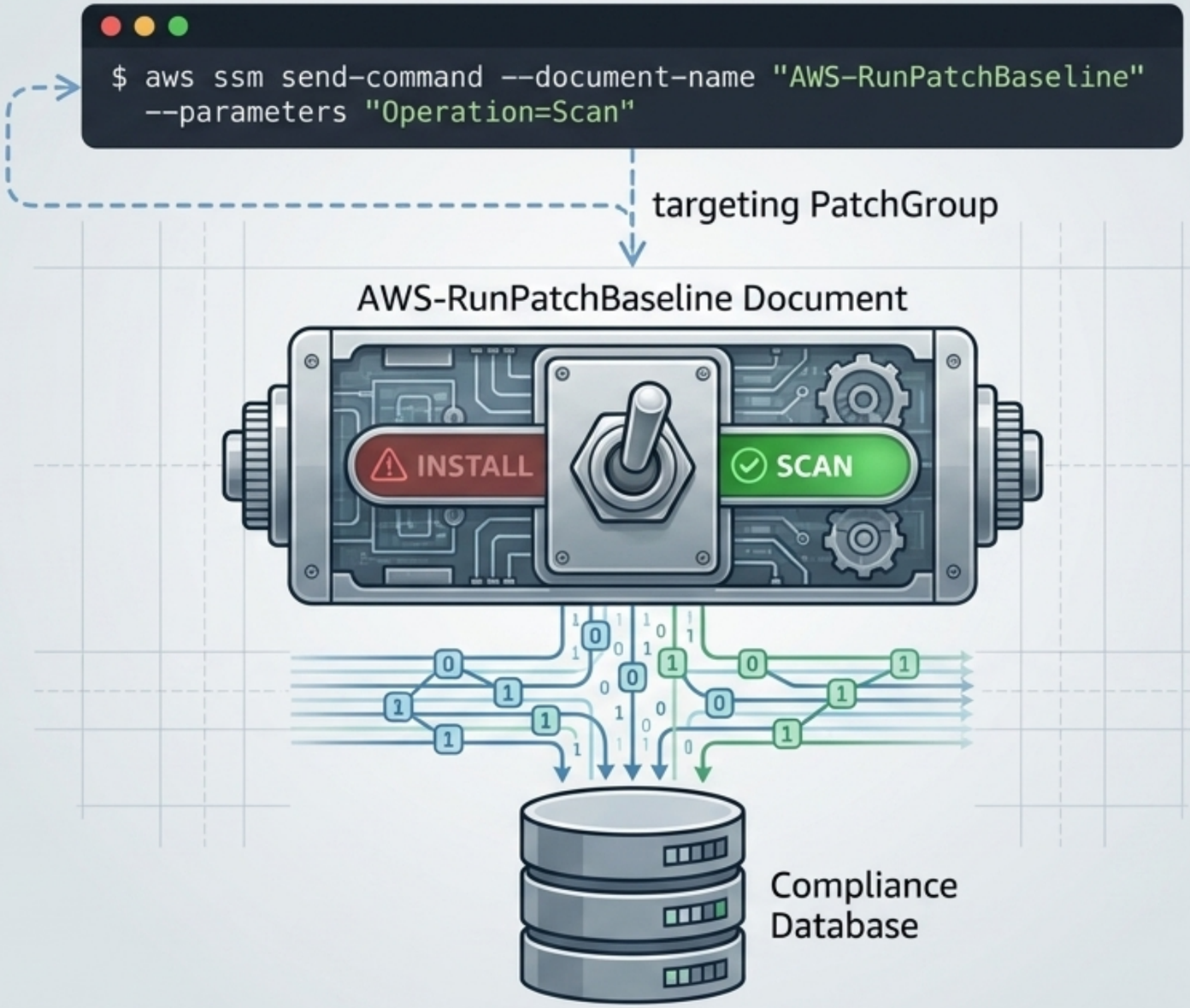
```
$ aws ec2 create-tags --resources ... --tags Key=PatchGroup,Value=Prod-Web
```

## Tagging Pipeline & Resource Sorting

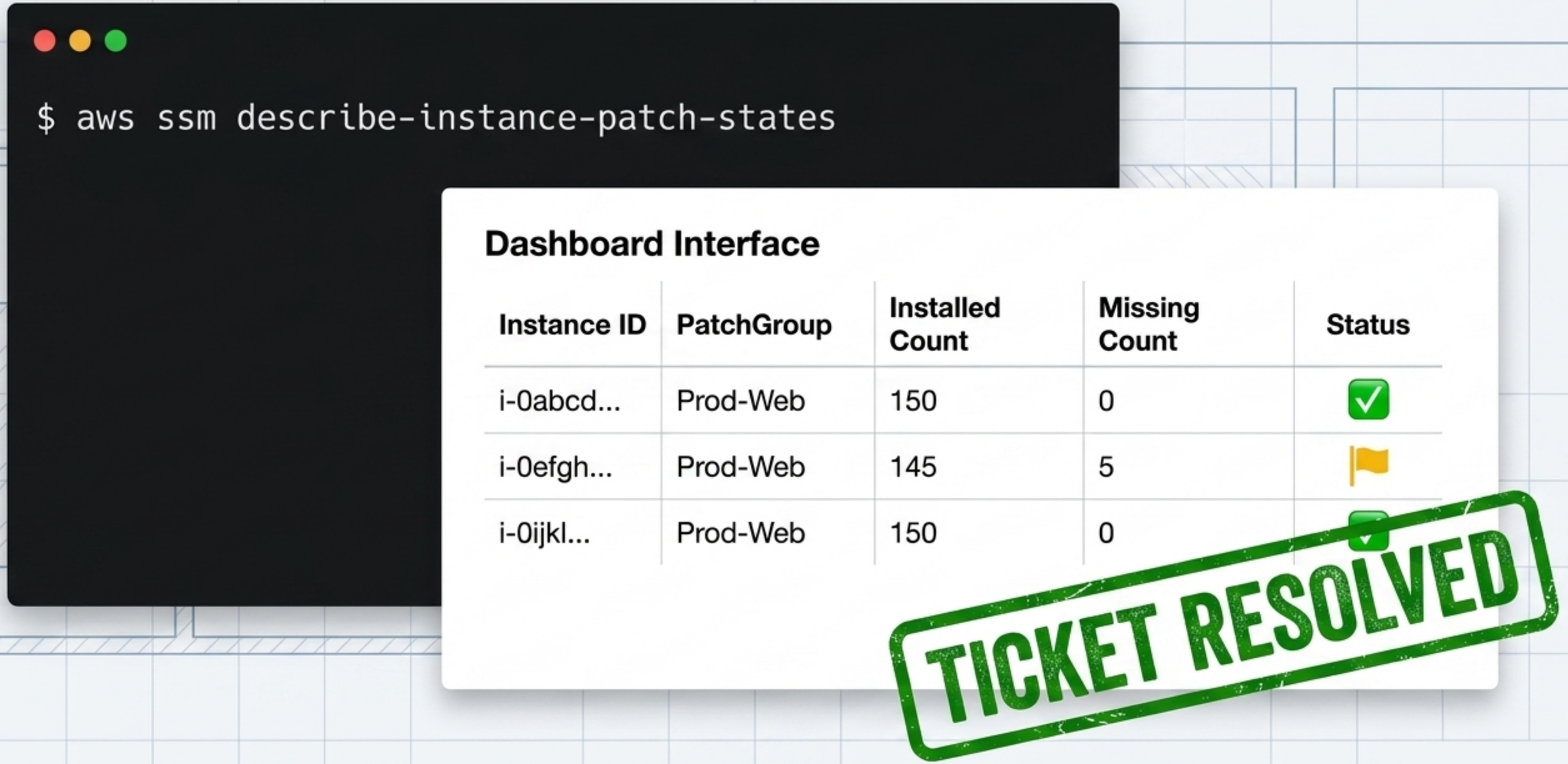



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# Executing the Read-Only Scan




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$ aws ssm describe-instance-patch-states
Dashboard Interface
Instance ID
PatchGroup
Installed Count
Missing Count
Status
i-0abcd...
Prod-Web
150
0
i-0efgh...
Prod-Web
145
5
i-0ijkl...
Prod-Web
150
0
TICKET RESOLVED

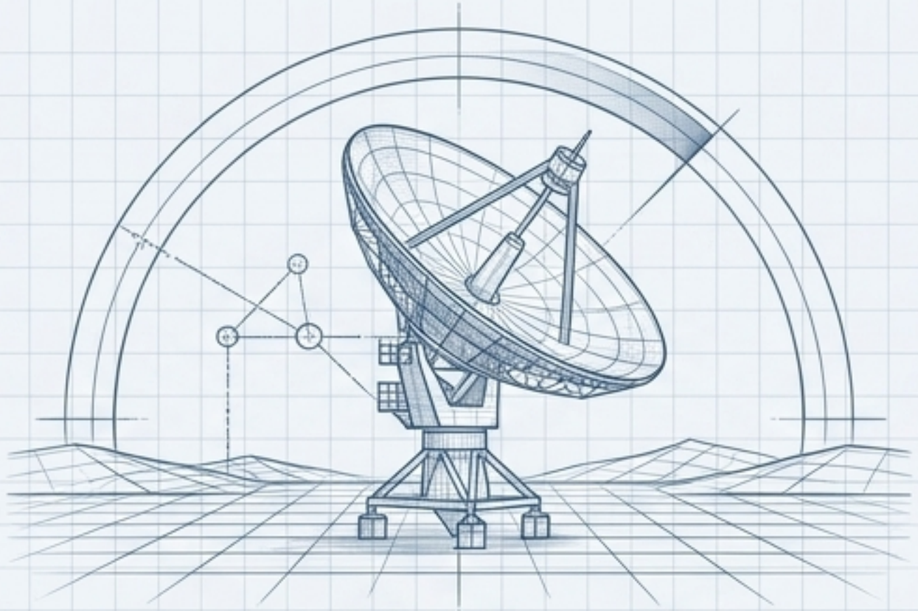
NotebookLM

# Patching Strategy Comparison: Cowboy vs. Enterprise

| | Cowboy Patching | Enterprise Change Control |
|---|---|---|
| Execution Trigger | `$ ssh user@instance -c "yum update -y"`<br>⚠ Ad-hoc, manual SSH execution | `$ aws ssm create-maintenance-window...`<br>✓ Scheduled, automated Maintenance Windows |
| Risk Profile | **High**<br>(Unpredictable downtime, fat-finger errors)  | **Low**<br>(Approved baselines, phased rollouts)  |
| Auditability | **Zero**<br>(No central record of who ran what)  | **Comprehensive**<br>(Compliance reporting, CloudTrail logged)   |
| Fleet Visibility | **Blind**<br>(Instance-by-instance guesswork)  | **Total**<br>(Centralized SSM dashboards)   |

Prod patching always runs inside a scheduled, approved window.

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# You can’t patch what you can’t manage.

| | |
|---|---|
| 1 | **Visibility First**: Always verify the SSM Triad (Agent, IAM, Network) before initiating fleet operations. |
| 2 | **Tag for Targeting**: Use strict metadata (PatchGroups) to separate logical environments (e.g., Prod vs. Non-Prod). |
| 3 | **Scan Before Install**: Always run read-only compliance scans to understand the state before mutating the environment. |
| 4 | **Control the Change**: Abandon the cowboy patch; enforce scheduled maintenance windows. |



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